RTM Imaging

Clay Wood

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Reverse Time Migration Background

- Time reversal relies on the concept of **reciprocity**: elastic wave equation is time-symmetric.
- Recorded seismograms from a point source can be reinjected (flipped in time) at the receiver locations. The back-propagating wavefield will focus in space–time at the original source coordinates.
- Model space: velocity model of subsurface.
- Data space: recorded seismic data.
- RTM determines best image of reflectors by applying an imaging condition to match the extrapolated time-reversed waveform data with the predictions from estimated velocity date ad source parameters.

Project Objective

Objective:

Use reverse time migration method to image small irregular reflectors in a subsurface velocity model.

This is an over-simplified toy model of a well-mated "fractured interface", where the reflectors have a lower velocity than the surrounding matrix and represent "porous" regions.

Initial Parameters and Boundary Conditions

- 2D Finite Difference Method
- Velocity Model: Figure 1
- Absorbing Boundary Conditions (bottom, left, right)

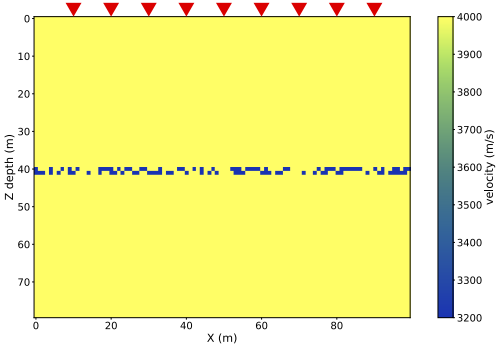


Figure 1: Velocity model with small irregular reflectors

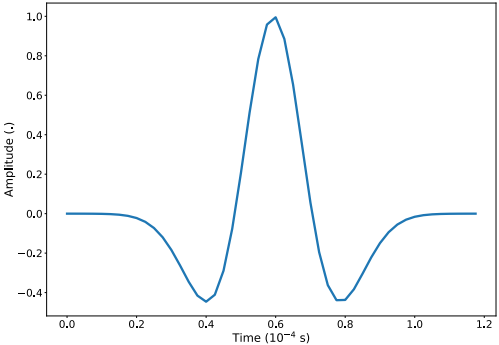
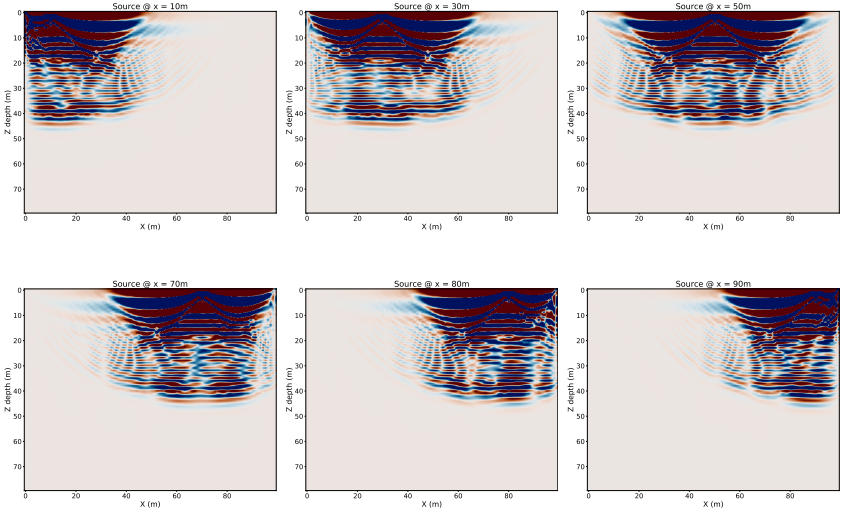


Figure 2: Excitation source: ricker wavelet with $f_c = 20\ kHz$

Down-going & up-going wavefields

Play .gif animations...

Shot-Gather: Pre-stack



Final Image

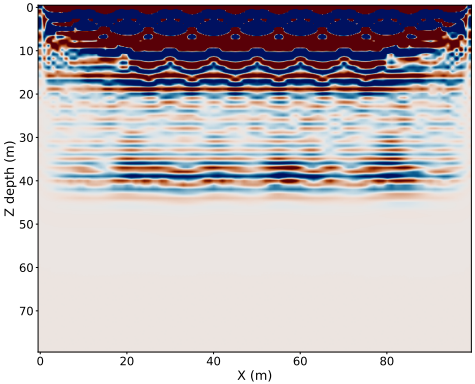


Figure 3: Stacked Image.

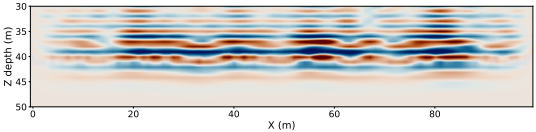


Figure 4: Final RTM image with overlayed velocity model.